Spin-1 Population Dynamics in Polarized Deuteron Targets: Mechanisms, Rate Equations, and Open Questions

Abstract

These notes collect the physics that governs the level populations of an ensemble of spin-1 nuclei (deuterons) in a dynamically polarized solid. They cover the Hamiltonian and powder spectrum, the reduction from the density matrix to rate equations, and every population-changing process relevant to polarized-target work:

- radio-frequency (RF) saturation and adiabatic fast passage (AFP);
- spin-lattice relaxation with separate vector and tensor channels;
- spectral spin diffusion with its three distinct spin-1 flip-flop channels;
- the nuclear dipolar reservoir (Provotorov theory);
- dynamic nuclear polarization by the solid effect, cross effect, thermal mixing and the differential solid effect.

The terms are assembled into one orientation- and site-resolved master equation, and we state its conservation laws and stationary states. Several general consequences for manipulating the tensor polarization follow directly from the structure of the equations. Throughout, we distinguish what is established theory, what is an effective closure, and what is not known for deuterated butanol or ammonia. Numbers quoted for d-butanol come from fits to UVA target data and are labelled as such.

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1 Scope, levels of description, and notation

The system is a frozen, amorphous solid (e.g. deuterated butanol C_4D_9OD or ND_3) at $T \simeq 1\,\mathrm{K}$ in $B_0 \simeq 2.5$–$5\,\mathrm{T}$, doped with paramagnetic centres (chemical radicals such as TEMPO or EHBA-Cr(V), or irradiation-induced radicals in ammonia) at concentration $c_e \sim 10^{19}$–$10^{20}\,\mathrm{cm}^{-3}$ [23, 24, 25]. The deuteron is a spin $I = 1$ nucleus with gyromagnetic ratio $\gamma_d/2\pi = 6.536\,\mathrm{MHz/T}$, so $\omega_d/2\pi \simeq 32.7\,\mathrm{MHz}$ at $5\,\mathrm{T}$. Its electric quadrupole moment couples to the electric-field gradient (EFG) of the chemical bond [4, 1, 3].

Three levels of description.

1. *Density matrix* of the coupled electron–nuclear system, with relaxation superoperators (Redfield theory) [1, 2]. This level is exact but intractable for $\sim 10^{22}$ spins.

2. *Spin-temperature (thermodynamic) theory*: weakly coupled reservoirs (electron Zeeman, electron dipolar, nuclear Zeeman, nuclear dipolar/quadrupolar), each internally equilibrated at its own inverse temperature [9, 10, 11, 12].

3. *Population (rate) equations* for the diagonal elements of the density matrix of spectrally resolved nuclear "packets". Coherences are adiabatically eliminated, and every process becomes a transition rate or a pair-exchange current. These notes work at this level and state explicitly where it fails.

Notation.

- Levels are $m \in \{+1, 0, -1\}$. For a deuteron ($\gamma_d > 0$) with $\mathcal{H}_Z = -\hbar\omega_d I_z$, the level $m = +1$ is lowest.
- Populations of the spins at orientation Ω (EFG principal axes relative to $\mathbf{B}_0$, polar angle θ, azimuth ϕ) and chemical site s are $n_m(\Omega, s, t)$.
- Fractional populations within a packet are p_m, with $p_{+1} + p_0 + p_{-1} = 1$.
- Powder averages are written $\langle f \rangle = \frac{1}{4\pi}\int f\, d\Omega$; if f is independent of ϕ this reduces to $\frac{1}{2}\int_0^\pi f \sin\theta\, d\theta$.

Polarizations.

$$P = \langle n_{+1} - n_{-1} \rangle\,, \qquad Q = \langle n_{+1} + n_{-1} - 2n_0 \rangle = 1 - 3\,\langle n_0 \rangle\,. \tag{1}$$

Positivity of the populations restricts (P, Q) to the triangle with vertices $(\pm 1, 1)$ and $(0, -2)$. A Boltzmann distribution with Zeeman inverse spin temperature β gives $n_m \propto e^{mx}$ with $x = \beta\hbar\omega_d$, so that

$$P = \frac{2\sinh x}{1 + 2\cosh x}, \qquad Q = \frac{2\cosh x - 2}{1 + 2\cosh x} = 2 - \sqrt{4 - 3P^2}. \tag{2}$$

Corrections of order ω_Q/ω_d are discussed in Sec. 6.3. At $P = 0.50$, Eq. (2) gives $Q = 0.1972$ and $(p_{+1}, p_0, p_{-1}) = (0.616, 0.268, 0.116)$; these values are used for illustrations throughout.

2 Hamiltonian, transition frequencies and the powder spectrum

2.1 Zeeman and quadrupolar interactions

In its principal-axis system the quadrupolar Hamiltonian is [4, 1, 3]

$$\mathcal{H}_Q = \frac{e^2qQ}{4I(2I-1)}\Big[3I_{z'}^2 - I(I+1) + \eta\,(I_{x'}^2 - I_{y'}^2)\Big], \tag{3}$$

where e^2qQ/h is the quadrupolar coupling constant and $0 \le \eta \le 1$ the EFG asymmetry. For aliphatic C–D bonds $e^2qQ/h \approx 160$–$170\,\mathrm{kHz}$ with $\eta \lesssim 0.05$; for hydrogen-bonded O–D, typically 200–230 kHz [3].

In high field, $|\mathcal{H}_Q| \ll \hbar\omega_d$. To first order only the part commuting with I_z survives:

$$\mathcal{H}_Q^{(1)} = \frac{e^2qQ}{4}\, P_2^\eta(\Omega)\,\big(3I_z^2 - 2\big) \qquad (I=1), \qquad P_2^\eta(\Omega) \equiv \tfrac{1}{2}\big(3\cos^2\theta - 1 + \eta\sin^2\theta\cos 2\phi\big). \tag{4}$$

The first-order energies are $E_m = -\hbar\omega_d m + \frac{e^2qQ}{4}P_2^\eta(3m^2-2)$, and the two allowed ($\Delta m = \pm 1$) transitions are

$$\boxed{\omega_+(\Omega) \equiv \omega_{+1\leftrightarrow 0} = \omega_d - \omega_Q\, P_2^\eta(\Omega), \qquad \omega_-(\Omega) \equiv \omega_{0\leftrightarrow -1} = \omega_d + \omega_Q\, P_2^\eta(\Omega), \qquad \omega_Q \equiv \frac{3e^2qQ}{4\hbar}.} \tag{5}$$

Convention. With this definition:

- the separation of the two transitions at one orientation is $2\omega_Q|P_2^\eta|$;
- the "horns" ($\theta = 90°$, $\eta = 0$) lie at offsets $\pm\omega_Q/2$, and the outer "shoulders" ($\theta = 0$) at $\pm\omega_Q$;
- both lines coincide at the magic angle ($\cos^2\theta = 1/3$).

The two forms in the earlier draft differ by a factor of 6 in the definition of ω_Q; Eq. (5) is used everywhere below. For the d-butanol C–D site the fitted horn offset is $\omega_Q/4\pi = 61.7\,\text{kHz}$, so $e^2qQ/h = 164\,\text{kHz}$.

2.2 Second-order and smaller terms

- **Second-order quadrupole shift.** For $I = 1$ it is of order $\omega_Q^2/\omega_d \sim 0.5\,\text{kHz}$ at 5 T for C–D. It shifts *both* transitions by the same orientation-dependent amount, so it does not change their splitting [4, 3]. It is below the homogeneous width but not negligible for O–D.
- **Chemical-shift anisotropy** of ^{2}H is a few ppm (about 0.1 kHz): negligible.
- **D–D dipolar couplings** are of order 0.1–1 kHz and set the homogeneous width (Sec. 2.3). D–H couplings matter if residual protons are present.
- **Hyperfine couplings** to the electrons shift deuterons within ∼1 nm of a radical far out of the bulk line (Sec. 7.4).

2.3 Homogeneous broadening

Each packet has a homogeneous width $\Gamma = 1/T_2$. It sets three things: the RF response (Sec. 4), the energy matching of flip-flops (Sec. 7), and the local line φ in Eq. (7). Contributions, roughly in order of size for deuterated glasses at 1 K:

1. **D–D dipolar couplings.** Van Vleck's second moment [6, 1] weights neighbours whose transitions are resonant ("like") 9/4 times more than non-resonant ("unlike") ones. For quadrupole-split deuterons, resonance depends on the neighbour's orientation, so Γ depends on the local density of matching transitions.
2. **Couplings to residual protons** (heteronuclear, always "unlike").
3. **Hyperfine fields of the paramagnetic centres.** These give a distribution of shifts, and they make the width position dependent (largest near radicals).

4. **Molecular motion** (methyl tunnelling, librations). This can narrow or broaden, depending on rate.

5. **Second-order quadrupole dispersion** for sites with a distribution of η or e^2qQ. This is inhomogeneous; it is included in the line fit.

The fitted common Lorentzian HWHM for d-butanol, 1.45 kHz, is an average. Its dependence on orientation and site is **not measured**, and it is the main physical candidate for per-spin RF selectivity (Sec. 4.3).

2.4 Pake density: the orientation Jacobian

For one site with $\eta = 0$, write the offset of a transition as $\delta = \omega - \omega_d = \mp\omega_Q P_2(\cos\theta)$. The fraction of spins per unit offset is the orientation Jacobian [5]:

$$g(\delta) = \frac{1}{|d\delta/d\cos\theta|} = \frac{1}{\sqrt{3}\,\omega_Q\sqrt{1+2\delta/\omega_Q}}, \qquad \delta \in [-\omega_Q/2,\, \omega_Q], \tag{6}$$

(for the branch whose offset is $+\omega_Q P_2$; the other branch is mirrored).

- The density diverges at the horn ($\theta = 90°$) and falls to its edge value at $\theta = 0$.
- With $\eta \neq 0$ the three principal values of P_2^η are 1 and $-(1 \pm \eta)/2$. The divergence moves to $\delta = -(1-\eta)\omega_Q/2$, a new step edge appears at $-(1+\eta)\omega_Q/2$, and the far edge stays at ω_Q [4, 3].
- Homogeneous broadening convolves each branch with the local line φ (Lorentzian or Voigt). An analytic complex form of the broadened branch is given in Ref. [26].

Throughout, the *capacity* μ of a spectral packet is the number of spins whose transition lies in that packet:

$$\mu(\delta)\,d\delta = (\text{number of spins}) \times g(\delta)\,d\delta.$$

The Jacobian is a density of states. It counts spins; it does not alter the physics of any single spin. This distinction matters for the RF rates (Sec. 4).

2.5 Several sites

A molecule such as C_4D_9OD contains several deuteron sites. Each has its own e^2qQ, η and dynamics:

- CD_2 and CD_3 groups;
- one O–D.

The spectrum is a weighted sum of Pake patterns. If a methyl group rotates rapidly, its effective coupling is scaled by $P_2(\cos 109.5°) = -1/3$. At 1 K, methyl rotation is frozen or proceeds only by tunnelling, and its effect on the line and on relaxation in d-butanol is not established.

A fit of 2014 UVA d-butanol spectra with a two-site (C–D, O–D) azimuth-averaged model gave:

- O–D/C–D coupling ratio 1.35;
- $\eta_{CD} = 0.065$ and $\eta_{OD} = 0.033$;

- an effective O–D integrated weight of 2.1%;
- a common Lorentzian HWHM of 1.45 kHz;
- held-out RMS of 0.79% of peak-to-peak.

A single-site model gave 0.80%. The second site is therefore only weakly required, and its weight should not be read as a chemical abundance.

2.6 The two overlapping lines

At any frequency in the spectrum two different groups of spins absorb:

- the I_+ transition ($+1 \leftrightarrow 0$) of spins at orientation Ω_1;
- the I_- transition ($0 \leftrightarrow -1$) of spins at the "mirror" orientation Ω_2, with $P_2^{\eta}(\Omega_2) = -P_2^{\eta}(\Omega_1)$.

These are different molecules with different capacities. At the negative horn, for example, the I_- transition belongs to the dense $\theta = 90^\circ$ spins, while the overlapping I_+ belongs to sparse spins near $\theta \approx 37^\circ$.

Every manipulation of Q is constrained by this overlap (Sec. 11).

2.7 Observables

The absorptive NMR signal is

$$I(\omega, t) = \sum_s \left\langle \big(n_{+1} - n_0\big)\varphi(\omega - \omega_+) + \big(n_0 - n_{-1}\big)\varphi(\omega - \omega_-) \right\rangle_s , \tag{7}$$

with an overall sign set by the detector phase (the earlier draft used the opposite sign convention). Integrating over each line separately,

$$A_+ = \langle n_{+1} - n_0 \rangle , \qquad A_- = \langle n_0 - n_{-1} \rangle , \qquad A_+ + A_- = P, \qquad A_+ - A_- = Q.$$

- The total area measures P.
- The area difference of the two branches measures Q.
- In equilibrium the peak-height ratio $r = I_+/I_- = e^x$ gives P through Eq. (2); this is the basis of the asymmetry method [26].

After a manipulation the populations are no longer Boltzmann, so the ratio method fails. In that case P and Q must be obtained from full line-shape fits.

3 From the density matrix to rate equations

3.1 Elimination of coherences

Take a pair of levels a, b connected by an off-diagonal perturbation V (an RF field, or the mixing term of a forbidden transition), with detuning Δ and coherence decay $1/T_2$. The coherence obeys

$$\dot{c}_{ab} = -(1/T_2 + i\Delta)c_{ab} + iV(n_a - n_b).$$

When $|V|T_2 \ll 1$, or more generally when the coherence decays faster than the populations change, c_{ab} follows the populations adiabatically. Substituting gives [1, 2]

$$\dot{n}_a = -W(n_a - n_b), \qquad W = \frac{2|V|^2 T_2}{1 + \Delta^2 T_2^2} = 2\pi |V|^2 g(\Delta), \tag{8}$$

where g is the normalized homogeneous line. This is Fermi's golden rule with a Lorentzian line. For a non-Lorentzian homogeneous line, g is replaced by the appropriate (e.g. Gaussian) shape function.

3.2 Validity

The rate description requires all of the following:

1. **Weak drive**, $\omega_1 T_2 \lesssim 1$, or at least that population changes be slow compared with T_2.
2. **Phase-averaged carriers**: no coherent interference between simultaneously applied tones.
3. **Secular approximation**: transitions separated by more than their widths are treated independently.

When condition 1 fails:

- a resonant pulse produces nutation (at $\sqrt{2}\,\omega_1$ for a single spin-1 transition) and spin locking;
- off-resonance irradiation is described by Redfield's rotating-frame thermodynamics [8] or Provotorov's two-temperature theory [9] (Sec. 4.5);
- quadrupolar echoes and multiple-quantum coherences require the full spin-1 density matrix [3].

3.3 Packet representation

Group spins into packets labelled by site s and by the frequency R of their I_+ transition. Their I_- transition then lies at the mirror frequency.

- The state is $n_m(R, s, t) = \mu_s(R)\, p_m(R, s, t)$.
- All spins in a packet share the same transition frequencies.
- Packets are therefore the natural variables: orientation enters only through the capacity $\mu_s(R)$ and through any orientation dependence of the rate coefficients (T_2, couplings).

Spins with different azimuth ϕ but the same frequency are lumped together when $\eta \neq 0$. This is exact for every process that depends only on frequency. It is an approximation for processes that depend on the full orientation, such as spatial correlations of neighbours.

4 RF saturation (ss-RF) and per-spin selectivity

Selective semi-saturating RF (ss-RF) burns spectral holes in the deuteron line to manipulate the tensor polarization [30, 31].

4.1 Weak-drive rates for spin 1

A linearly polarized RF field contains a rotating component of amplitude B_1, with $\omega_1 = \gamma_d B_1$. In the rotating frame $V = \omega_1 I_x$.

Matrix elements. For spin 1,

$$|\langle +1|I_x|0\rangle|^2 = |\langle 0|I_x|-1\rangle|^2 = \tfrac{1}{2}.$$

To first order in ω_Q/ω_d these do not depend on orientation, because the quadrupole interaction does not mix the Zeeman eigenstates at that order.

Rate. From Eq. (8), a carrier at ω_c drives a transition at $\omega_\pm$ with

$$W_\pm = \frac{\omega_1^2 T_2}{1 + (\omega_\pm - \omega_c)^2 T_2^2}, \qquad J_\pm = W_\pm \left(n_{\text{lower}} - n_{\text{upper}}\right). \tag{9}$$

- A carrier exactly on resonance with a two-level pair equalizes it at rate $2W$; with lattice relaxation the steady state is $d = d_0/(1 + 2WT_1)$.
- Independent carriers (random relative phases) add in power: $W = \sum_c W^{(c)}$.
- RF power is proportional to B_1^2, so the rates are linear in power.

4.2 Spin density does not enter the rate per spin

Equation (9) contains ω_1, the detuning and T_2. It does *not* contain how many spins share the frequency $\omega_\pm$:

- the Pake Jacobian (the horn) puts more spins per hertz near $\theta = 90°$;
- each of those spins sees the same B_1 and has the same matrix element;
- B_1 is a field, not a resource shared among the spins: at the RF powers used, spin absorption is a negligible load on the coil.

Hence the absolute rate of polarization loss of a packet is proportional to its capacity μ (through $n = \mu p$), but its *fractional* saturation rate is not. A phenomenological rate proportional to the powder density (used in early models) counts the spins twice.

4.3 Physical sources of per-spin selectivity

Selectivity means that the two overlapping transitions at one frequency (Sec. 2.6) saturate at different fractional rates. It can only arise from differences in ω_1, the matrix element, or T_2 between the two groups of spins, or from processes beyond single-spin saturation.

1. **Orientation- and site-dependent T_2.** On resonance, $W \propto T_2$ at fixed ω_1. The homogeneous width comes from D–D and D–H dipolar couplings, from proximity to paramagnetic centres, and from molecular motion. All of these depend on molecular orientation and site (Sec. 2.3). This is the most direct physical route to selectivity, and it can be measured from narrow-hole widths.

2. **The like-spin second moment.** In Van Vleck's theory [6, 1], a dipolar neighbour contributes 9/4 times more to the second moment when it is "like" (resonant within the linewidth) than when it is "unlike". For quadrupole-split deuterons, a neighbour counts as like only if its transition matches within the homogeneous width. The homogeneous width at a frequency therefore grows with the local density of matching transitions, by up to $(9/4)^{1/2} = 1.5$ in width. This is a genuine density-dependent per-spin effect. Its sign is to *broaden* dense regions and so to lower their on-resonance rate.

3. **Spatial B_1 inhomogeneity.** The coil, the target cup and the packing give a distribution of ω_1 across the sample. It is uncorrelated with frequency, so it produces no selectivity. It does produce a slow, non-exponential tail of saturation (a candidate contributor to apparent plateaus, Sec. 12).

4. **Hyperfine-shifted spins** near radicals are detuned and saturate differently (Sec. 7.4).

5. **Collective effects.** Radiation damping and RF-induced changes of coil loading depend on the magnetization at a frequency. For deuterons at the signal levels of polarized targets they are expected to be small.

Status: *Eq. (9) is established for weak drive. The orientation dependence of T_2 in d-butanol and ND_3 is* ***not measured****. No first-principles mechanism makes the rate per spin proportional to powder density.*

4.4 Strong drive, rotating-frame relaxation and coherent effects

When $\omega_1 \gtrsim$ the local dipolar field, Eq. (8) breaks down:

- **Off resonance:** the spin system reaches a rotating-frame spin temperature [8, 10].
- **On resonance:** spin locking preserves the magnetization against saturation, which then decays with $T_{1\rho}$.
- **Pulsed operation:** nutation and quadrupolar echoes, which can invert populations rather than equalize them (AFP is the adiabatic limit, Sec. 5).

In polarized-target practice the RF level is set by the Q-meter circuit. Whether $\omega_1 T_2 < 1$ holds must be checked against a measured B_1 calibration.

4.5 Coupling of off-resonance RF to the nuclear dipolar reservoir

Off-resonance RF transfers energy between the Zeeman system and the nuclear dipolar (and quadrupolar non-Zeeman) reservoir [9, 10].

Single homogeneous line. Take a line at offset Δ from the RF, with rotating-frame Zeeman inverse temperature α, dipolar inverse temperature β and local field D (in frequency units). Provotorov's equations are

$$\dot{\alpha} = -W(\Delta)(\alpha - \beta) - \frac{\alpha - \alpha_L}{T_{1Z}}, \qquad \dot{\beta} = -W(\Delta)\frac{\Delta^2}{D^2}(\beta - \alpha) - \frac{\beta - \beta_L}{T_{1D}}. \tag{10}$$

Packet form. For an inhomogeneous (Pake) line at the population level, take a carrier at ω_c driving a transition at ω. The current acquires a term that is first order in the dipolar inverse temperature β_d:

$$J = W\Big[(n_{\text{lower}} - n_{\text{upper}}) - \hbar\beta_d(\omega - \omega_c)\,\tfrac{1}{2}(n_{\text{lower}} + n_{\text{upper}})\Big], \qquad C_d\,\dot{\beta}_d = \sum(\omega - \omega_c)J + \dots \tag{11}$$

- Absorption below resonance takes the energy mismatch from the reservoir and cools it.
- This is the nuclear analogue of dynamic cooling. It also makes RF hole shapes asymmetric.

Status: *The theory is established [9, 10, 12]. The heat capacity C_d of the deuteron non-Zeeman reservoir in d-butanol, and its thermal contact with the quadrupole-split Zeeman packets, are* ***unknown****. The first-order form requires $\hbar\beta_d|\omega-\omega_c| \ll 1$.*

5 Adiabatic fast passage

A frequency or field sweep through a transition at rate $\dot\omega$ inverts it with Landau–Zener efficiency

$$\epsilon = 1-\exp\Big(-\frac{\pi\,\Omega_R^2}{2|\dot\omega|}\Big) = 1-\exp\Big(-\frac{\pi\,\omega_1^2}{|\dot\omega|}\Big), \qquad \Omega_R = 2\,|\langle a|\omega_1 I_x|b\rangle| = \sqrt{2}\,\omega_1, \tag{12}$$

where Ω_R is the Rabi frequency of the isolated two-level transition (the other transition of the same spins is far off resonance). This holds provided the sweep is fast compared with $T_{1\rho}$ and slow compared with ω_1 [1, 2, 22].

Effect at one frequency. A sweep through frequency R inverts both transitions there. It swaps $n_{+1} \leftrightarrow n_0$ for the packet whose I_+ lies at R, and $n_0 \leftrightarrow n_{-1}$ for its mirror packet.

Gain in Q. Using $Q = 1-3\,\langle n_0\rangle$, the change from inverting at R is

$$\Delta Q_{\mathrm{AFP}}(R) = 3\epsilon\Big[\mu_{\mathrm{m}}\big(p_0-p_{-1}\big)_{\mathrm{m}} - \mu_{\mathrm{i}}\big(p_{+1}-p_0\big)_{\mathrm{i}}\Big], \tag{13}$$

where i is the packet whose I_+ lies at R and m its mirror. The gain is positive where the mirror packet is more than $(p_{+1}-p_0)/(p_0-p_{-1}) \approx 2.3$ times denser (at $P = 0.5$).

Inversion moves twice the population that saturation does, so selective AFP windows can raise Q well beyond what selective saturation can reach. For d-butanol the beneficial windows are the negative horn region and the outer positive region beyond the C–D edge.

Limits of the idealization. The adiabatic condition, relaxation during the sweep, sweep edges, and a sweep passing through both transitions of the same spins (which gives a $+1 \leftrightarrow -1$ exchange) limit real efficiencies.

Status: *Physics established. The selective-window efficiencies achievable in a polarized-target Q-meter set-up are* ***not measured****.*

6 Spin-lattice relaxation

6.1 Quadrupolar relaxation of spin 1

Fluctuations of the EFG (molecular motion and phonons) induce $\Delta m = \pm 1$ transitions at rate W_1 and $\Delta m = \pm 2$ transitions at rate W_2 [1, 3]. Linearizing about equilibrium,

$$\dot P = -(W_1+2W_2)(P-P_L), \qquad \dot Q = -3W_1(Q-Q_L), \tag{14}$$

with the rates defined per transition in each direction. Vector and tensor polarization therefore relax with *different* times,

$$T_{1Z}^{-1} = W_1+2W_2, \qquad T_{1Q}^{-1} = 3W_1,$$

and both depend on orientation through the spectral densities.

6.2 Paramagnetic relaxation

In doped targets, nuclear relaxation is dominated by the radicals. Nuclei near a centre relax directly through the fluctuating hyperfine field [7, 15]. The bulk relaxes by spin diffusion towards those centres, in the diffusion-limited or rapid-diffusion regimes of Khutsishvili and Blumberg, which gives $T_1^{-1} \propto c_e$ [15, 11]. A treatment specific to polarized targets is given in [28]. Both mechanisms are $\Delta m = \pm 1$ dominated.

6.3 Lattice and spin-temperature equilibria

At the lattice temperature $T_L \simeq 1\,\mathrm{K}$, $P_L \approx 10^{-3}$. Under DNP, the nuclear Zeeman system instead approaches a spin temperature set by the electrons (Sec. 9).

Quadrupole corrections. The full Boltzmann distribution includes the quadrupolar energy, $n_m \propto \exp[-\beta E_m]$. This gives orientation-dependent corrections to Eq. (2) of relative order $\beta\hbar\omega_Q$. At deuteron spin temperatures corresponding to $P \simeq 0.5$ these are about 0.4%.

Separate quadrupolar order. The tensor (quadrupolar-order) part can acquire its own temperature after a manipulation. It returns to the value set by P only through processes that change n_0: lattice relaxation ($3W_1$), cross-branch spin flips (Sec. 7), and DNP.

Status: *Mechanisms established. Orientation dependence, the W_2/W_1 ratio and the absolute T_1 of deuterons in d-butanol at 1 K are* ***not separately measured****; in practice T_1 is hours to days.*

7 Spectral spin diffusion for spin 1

7.1 Dipolar flip-flops between quadrupole-split spins

Two deuterons i, j couple through the secular dipolar Hamiltonian [1, 7]

$$\mathcal{H}_{ij} = b_{ij}\big[2I_{zi}I_{zj} - \tfrac{1}{2}(I_{+i}I_{-j} + I_{-i}I_{+j})\big], \qquad b_{ij} = \frac{\mu_0\gamma_d^2\hbar}{4\pi r_{ij}^3}\,\tfrac{1}{2}(1 - 3\cos^2\vartheta_{ij}),$$

where ϑ_{ij} is the angle of the internuclear vector to $\mathbf{B}_0$. For spin 1,

$$\langle +1|I_+|0\rangle = \langle 0|I_+|-1\rangle = \sqrt{2}.$$

The flip-flop term therefore connects the following pair states, each with a matrix element of magnitude $|b_{ij}|$:

Channel	Pair transition	Energy matched when
same-branch I_+	$\lvert 0_i, +1_j\rangle \leftrightarrow \lvert +1_i, 0_j\rangle$	$\omega_+(\Omega_i) = \omega_+(\Omega_j)$
same-branch I_-	$\lvert -1_i, 0_j\rangle \leftrightarrow \lvert 0_i, -1_j\rangle$	$\omega_-(\Omega_i) = \omega_-(\Omega_j)$
cross-branch	$\lvert 0_i, 0_j\rangle \leftrightarrow \lvert +1_i, -1_j\rangle$	$\omega_+(\Omega_i) = \omega_-(\Omega_j)$

Same matrix element, different partners. All three channels have the same dipolar matrix element. The cross-branch process exchanges excitation between the two *overlapping* lines at one spectral position: it couples the I_+ of one spin with the I_- of a spin at the mirror orientation. In a random powder its strength relative to the same-branch channels is set by

- the density of energy-matched partners: same-branch partners at frequency R have capacity $\mu(R)$, cross-branch partners the mirror capacity;

- any orientation correlation between dipolar-coupled neighbours (intramolecular deuterons have fixed relative bond orientations).

A cross-branch/same-branch coupling ratio of order unity is therefore the *expected* value for uncorrelated pairs, not an arbitrary choice.

Double-quantum channel. The exchange $|+1_i, -1_j\rangle \leftrightarrow |-1_i, +1_j\rangle$ needs two flip-flops. To first order it conserves both Zeeman and quadrupolar energy for *any* orientations, because E_{+1} and E_{-1} have the same quadrupole shift. It is therefore nonlocal in the spectrum. Through an off-resonant intermediate $|0, 0\rangle$, its amplitude is of order b_{ij}^2/δ, where δ is the intermediate energy mismatch, so its rate is suppressed by roughly $(b/\delta)^2$ relative to resonant flip-flops.

Quadrupolar quenching. Flip-flops between spins whose transitions differ by more than the homogeneous width are truncated. Spectral diffusion in resolved quadrupolar spectra is therefore local in frequency and channel-specific [17]. Energy mismatches can be bridged by the nuclear dipolar reservoir (three-spin processes, Sec. 8).

7.2 Pair-current rate equations

Let $S(\Delta)$ be the zero-quantum spectral overlap of the two transitions involved (width of order the homogeneous width or the dipolar local field), and let C_{ij} describe spatial and orientation correlation. The packet equations are reversible pair currents.

Same-branch (I_+ shown; I_- analogous):

$$J_{ij}^{+} = K\,\mu_i\mu_j\,C_{ij}\,S(R_i - R_j)\big(p_{0i}p_{+j} - p_{+i}p_{0j}\big). \tag{15}$$

Cross-branch (the I_+ of i at R_i with the I_- of j at $-R_j$):

$$J_{ij}^{X} = Kr_X\,\mu_i\mu_j\,C_{ij}\,S(R_i + R_j)\big(p_{0i}p_{0j} - p_{+i}p_{-j}\big). \tag{16}$$

Double-quantum:

$$J_{ij}^{DQ} = Kr_{DQ}\,\mu_i\mu_j\,C_{ij}\big(p_{-i}p_{+j} - p_{+i}p_{-j}\big). \tag{17}$$

Properties [17, 12]:

- All currents conserve packet mass and total P.
- Same-branch currents conserve each level total separately.
- Cross-branch currents change Q at fixed P; they are a spin-spin route by which tensor order relaxes to the value set by P.
- Double-quantum currents move vector polarization across the spectrum without changing Q.
- Every current vanishes for a common Boltzmann distribution. The stationary state is unique when the exchange graph is connected.

The currents are bilinear in the populations. Spin diffusion therefore slows where partner transitions are saturated ("flip-flop blocking"), and it cannot be represented as each packet relaxing to a fixed reference.

7.3 Continuum limit: diffusion along the Pake line

For small spectral jumps, the kernel reduces to a diffusion equation in frequency. Linearizing for a single branch with $\mu(\omega)$ the Pake density,

$$\partial_t p = \frac{1}{\mu(\omega)}\,\partial_\omega\big[\mu(\omega)\,\tilde{D}(\omega)\,\partial_\omega p\big], \qquad \mu(\omega) \propto \frac{\sin\theta}{|d\omega/d\theta|}. \tag{18}$$

This is the earlier draft's Laplace–Beltrami operator $\sin^{-1}\theta\,\partial_\theta(D\sin\theta\,\partial_\theta n)$ rewritten with the Jacobian $J = |d\omega/d\theta| = \Omega\sqrt{2(1-x)(1+2x)}$, $x = \omega/\Omega$, and $\tilde{D} = DJ^2$.

Three caveats apply:

1. The variable θ labels packets. Spins do not reorient; "diffusion in θ" describes energy-conserving exchange between molecules.
2. Cross-branch and double-quantum transport are nonlocal and *cannot* be written in this form.
3. The no-flux boundary conditions apply at the edges of the support, $\omega \in \{-\Omega/2, \Omega\}$.

7.4 Real space, diffusion barrier and microwaves

Spectral transport requires spatial proximity, $|b_{ij}| \propto r_{ij}^{-3}$. A Pake histogram does not specify which spectral packets are spatial neighbours: the correlation C_{ij} is an effective closure.

Diffusion barrier. Deuterons within the "diffusion barrier" of a radical, where the hyperfine shift exceeds the dipolar coupling, are detuned from the bulk and exchange with it only weakly [16, 11, 21].

Microwave effects. Microwave irradiation can modify nuclear spin diffusion by reducing the electron polarization. Electron-polarization-dependent proton spin diffusion has been observed at 1.2 K [29].

Status: *The mechanisms are established. The deuteron K, S width, r_X, r_{DQ} and correlations in d-butanol are* ***weakly identified****. A fit to one burn/refill experiment gave a nonlocal channel much stronger than the $(b/\delta)^2$ estimate. That channel is therefore effective, and may absorb reservoir-assisted or real-space transport.*

8 Thermal mixing among the nuclei

Flip-flops between packets of different frequency are allowed when the energy mismatch is supplied or absorbed by the nuclear dipolar reservoir (three-spin processes) [10, 12]. With reservoir inverse temperature β_d, detailed balance gives, to first order,

$$J_{ij} = H_{ij}\Big[(p_{0i}p_{+j} - p_{+i}p_{0j}) + \tfrac{1}{2}\hbar\beta_d\,E_{ij}\,(p_{0i}p_{+j} + p_{+i}p_{0j})\Big], \tag{19}$$

where E_{ij} is the energy released. Analogous terms apply to the cross-branch channel, with $E = \hbar(R_i + R_j)$. The double-quantum channel releases no energy at first order.

Energy conservation. The spin offset energy $E_{\text{spin}} = -\hbar\sum R(n_{+1} + n_{-1})$ and the reservoir energy $E_d = -C_d\beta_d$ are conserved together:

$$\dot{E}_{\text{spin}} + \dot{E}_d = 0.$$

Thermal mixing drives the whole line towards a common rotating-frame spin temperature. It broadens the effective exchange width beyond the homogeneous width, and it couples to RF through Eq. (11).

Status: *Theory established. Its importance for deuterons in d-butanol, and C_d, are* ***unknown****.*

9 Dynamic nuclear polarization of spin-1 nuclei

9.1 Overview and regimes

Microwaves at ω_μ near the electron Larmor frequency ω_e (140 GHz at 5 T) transfer electron polarization to the nuclei. Which mechanism dominates depends on the ESR width Δ_e relative to ω_d, on electron spectral diffusion and on c_e [13, 11, 18, 21]:

Regime	Condition	Mechanism
Solid effect (SE)	$\Delta_e \ll \omega_d$	forbidden e–n flips at $\omega_e \pm \omega_n$
Differential SE (DSE)	$\Delta_e \gtrsim \omega_d$, weak e–e spectral diffusion	overlapping SE sidebands, net difference
Cross effect (CE)	$\Delta_e > \omega_d$, e–e coupled pairs	three-spin e_1e_2n with $\lvert\omega_{e1} - \omega_{e2}\rvert = \omega_n$
Thermal mixing (TM)	$\Delta_e > \omega_d$, fast e–e spectral diffusion	spin-temperature contact with electron dipolar reservoir

For d-butanol with TEMPO or EHBA-Cr(V) at 5 T, Δ_e is several hundred MHz, far larger than $\omega_d/2\pi = 32.7$ MHz. DNP is then usually described by spin-temperature theory (TM), with DSE contributions discussed for ammonia [24, 25, 18, 21].

For spin 1 every mechanism acts *separately* on the two transitions of each orientation, at their own frequencies $\omega_\pm(\Omega)$.

9.2 Solid effect

The hyperfine interaction mixes states, so microwaves at

$$\omega_\mu \simeq \omega_e \mp \omega_\pm(\Omega)$$

flip the electron and the nucleus together [13, 11, 19]. In the weak-mixing limit the effective operator is

$$\mathcal{H}_{SE}^{\text{eff}} \propto (A_\perp/\omega_d)\,(S_+I_- + S_-I_+),$$

so the transition rate is proportional to $(A_\perp/\omega_d)^2\omega_{1e}^2$ times the electron spectral density. (The mixing coefficient involves ω_n in the denominator, not ω_e.)

For each branch $b \in \{+,-\}$, nucleus-raising ($\uparrow$) and lowering ($\downarrow$) rates are

$$W_b^{\uparrow,\downarrow}(\Omega;\omega_\mu) = k_{SE}\,c_e\,\langle A_\perp^2\rangle\,\frac{\omega_{1e}^2 T_{2e}}{1+\Delta_\mp^2 T_{2e}^2 + \omega_{1e}^2 T_{1e}T_{2e}}\,L_e\big(\Delta_\mp\big), \qquad \Delta_\mp = \omega_\mu - \big[\omega_e \mp \omega_b(\Omega)\big]. \quad (20)$$

Here L_e is the ESR line and $\beta = \omega_{1e}^2 T_{1e}T_{2e}/(1+\Delta^2 T_{2e}^2)$ is Jeffries' pumping parameter.

The net pumping direction is proportional to the local electron polarization, $P_e^{\text{loc}} = P_e^{(0)}/(1+\beta)$. Double-quantum SE ($\Delta m_I = \pm 2$) is allowed only at higher order and is weak.

The spin-1 SE rate equations in the narrow-line limit, including phonon channels, are given in [27]. The orientation-resolved generalization adds three things:

- the full Ω dependence through $\omega_\pm(\Omega)$;
- arbitrary ESR shape;
- spin diffusion (Sec. 7).

9.3 Differential solid effect

When $\Delta_e \gtrsim \omega_d$, the positive and negative SE sidebands from different spin packets overlap. The net polarization comes from the *difference* between packets on the two sides of ω_μ [21]. The rates become integrals of Eq. (20) over the ESR line:

$$W_b^{\uparrow} - W_b^{\downarrow} \propto \int d\omega\, s_e(\omega - \omega_\mu)\, L_e(\omega) \left[L_e(\omega + \omega_b) - L_e(\omega - \omega_b)\right]. \tag{21}$$

The result is a smooth S-curve whose zero crossing is shifted by line asymmetry (g-anisotropy) and by saturation gradients.

9.4 Cross effect

Two dipolar-coupled electrons whose Larmor frequencies differ by a nuclear transition frequency,

$$|\omega_{e1} - \omega_{e2}| \simeq \omega_\pm(\Omega),$$

exchange polarization with the nucleus through the three-spin term $S_{1\pm}S_{2\mp}I_\pm$ [20, 21]. Microwaves saturate one electron of the pair. The rate scales with the pair density, and so with c_e^2:

$$W_b^{CE,\uparrow\downarrow} \propto c_e^2 \left\langle b_{ee}^2 \right\rangle \int d\omega\, s_e(\omega - \omega_\mu) L_e(\omega) L_e(\omega \mp \omega_b) \left[P_e(\omega \mp \omega_b) - P_e(\omega)\right]. \tag{22}$$

9.5 Thermal mixing and spin temperature

If electron spectral diffusion is fast, the electron non-Zeeman (dipolar plus offset) energy has a single inverse temperature β in the frame rotating at ω_μ [9, 14, 11, 10].

Homogeneous ESR line. With offset $\Delta = \omega_e - \omega_\mu$, Provotorov's equations (10) apply to the electrons. Under strong saturation ($\alpha = \beta$) the steady state is

$$\beta \simeq \beta_L \, \frac{\omega_e\, \Delta}{\Delta^2 + D_e^2\, T_{1Z}/T_{1D}}. \tag{23}$$

This is odd in Δ, with extrema at $|\Delta| = D_e\sqrt{T_{1Z}/T_{1D}}$, where D_e is the electron local field.

Inhomogeneous ESR line. Borghini's spin-temperature model gives the electron polarization profile across the line, and β is fixed by energy balance between microwave absorption and electron spin-lattice relaxation [14, 11].

Contact with the nuclei. Through three-spin e–e–n flips, the nuclear Zeeman system approaches the same β:

$$\dot{\boldsymbol{n}}(\Omega)\big|_{TM} = -\frac{1}{T_{IS}(\Omega)}\Big[\boldsymbol{n}(\Omega) - \boldsymbol{n}^{ST}(\beta;\Omega)\Big], \qquad n_{+1}^{ST} : n_0^{ST} : n_{-1}^{ST} = e^{\beta\hbar\omega_+} : 1 : e^{-\beta\hbar\omega_-}, \tag{24}$$

with each transition driven toward the ratio set by its own frequency. The thermal-contact time satisfies $T_{IS}^{-1} \propto c_e$ times the electron dipolar spectral density at the nuclear frequency.

Two consequences:

1. DNP by TM drives P and Q toward the spin-temperature relation (2), up to quadrupole corrections of order $\beta\hbar\omega_Q$. It acts as an *absolute* target, $\beta(\omega_\mu)$, not as relaxation toward the current P.

2. If the contact is stronger near the horns or for particular sites, DNP itself is orientation selective.

Heat balance. A consistent three-reservoir model (electron Zeeman, electron dipolar, nuclear Zeeman, inverse temperatures α, β, β_n) has the form

$$C_n\dot{\beta}_n = -\frac{C_n(\beta_n-\beta)}{T_{IS}} - \frac{C_n(\beta_n-\beta_L)}{T_{1n}}, \qquad C_D\dot{\beta} = \cdots + \frac{C_n(\beta_n-\beta)}{T_{IS}}, \tag{25}$$

where the back-reaction of the nuclei on the electron dipolar reservoir ("nuclear heat load") matters for high nuclear heat capacity.

Microwave on/off asymmetry. Microwaves can also change nuclear transport (by reducing the electron polarization) separately from pumping [29]. An observation that about three times more RF power is needed to reach the same peak residual with microwaves on constrains the sum of these effects. It does not separate them.

Status: *SE, CE, TM and DSE theory are established in the cited texts. For spin-1 deuterons in d-butanol the relative weight of the mechanisms, and their orientation dependence, are* ***not identified****. A phenomenological "relax to the spin-temperature target" form, with rate and saturation P fitted to build-up data, describes the measured 2014 build-up to 0.8% RMS.*

10 The complete master equation

Collect all processes for packet $k = (R, s)$. Its I_+ transition lies at R and its I_- at the mirror.

$$\dot{\boldsymbol{n}}_k = \underbrace{\boldsymbol{R}_{\mathrm{RF}}(\boldsymbol{U},\beta_d)\boldsymbol{n}_k}_{\text{Sec. 4}} + \underbrace{\boldsymbol{R}_{\mathrm{DNP}}\boldsymbol{n}_k + \boldsymbol{s}_k^{\mathrm{DNP}}}_{\text{Sec. 9}} + \underbrace{\boldsymbol{R}_{T_1}(\boldsymbol{n}_k - \boldsymbol{n}_k^L)}_{\text{Sec. 6}} + \underbrace{\sum_l \boldsymbol{J}_{kl}(\boldsymbol{n},\beta_d)}_{\text{Secs. 7,8}} \tag{26}$$

Here:

- $\boldsymbol{R}_{\mathrm{RF}}$ has rates $g_k^{\pm} = c_k \sum_j \kappa\Gamma_k/[\Gamma_k^2 + (\omega_k^{\pm} - \omega_j)^2]\, U_j$, where U_j is the carrier power, c_k the per-spin coupling and $\Gamma_k = 1/T_{2,k}$.
- $\boldsymbol{R}_{T_1}$ contains W_1 and W_2.
- $\boldsymbol{J}_{kl}$ are the three spin-diffusion channels, with the thermal-mixing corrections of Eq. (19).
- With the dipolar reservoir included, β_d obeys $C_d\dot{\beta}_d = \sum(\text{offset})\times(\text{current}) - C_d(\beta_d - \beta_L)/T_{1d}$.

AFP enters as an instantaneous map between integration intervals.

Conservation and balance.

- Every term preserves packet mass.
- Spin diffusion conserves P.
- Exchange together with the reservoir conserves offset energy.
- The RF terms remove P and drive towards local saturation.
- DNP and T_1 are the only sources that restore P.

Without RF, the unique stationary state is the spin-temperature state of the dominant source, modified by the reservoir.

Discretization. The spectral grid must resolve the narrowest of Γ_k, the exchange width and the line edges. Pair sums over Lorentzian tails are Toeplitz (or Hankel, for the cross-branch channel) and are evaluated efficiently by FFT. Forward-Euler integration must respect $h \sum(\text{outgoing rates}) < 1$.

11 General consequences for tensor polarization

The following results follow from Eqs. (1), (9) and (26). The numbers in brackets are for d-butanol at $P_0 = 0.50$ ($Q_0 = 0.197$); they come from numerical optimization in the fitted model, are model-dependent, and are given only as orders of magnitude.

1. **Saturation versus overlap.** Saturating the I_- transition of a packet with capacity μ raises Q by $\frac{3}{2}\mu(p_0 - p_{-1})$ [0.151μ]. Saturating an I_+ lowers it by $\frac{3}{2}\mu(p_{+1} - p_0)$ [0.349μ]. A saturating burn at one frequency always affects both overlapping transitions, so its net value depends on their capacities and on their relative per-spin rates.

2. **Equal per-spin rates.** The only lever left is the choice of frequencies. Optimized single burns reach [$Q \approx 0.237$].

3. **Perfect selectivity ceiling.** The two transitions of one packet share its coupling, so at most the denser packet of each mirror pair can be selectively saturated:
$$Q \le Q_0 + \tfrac{3}{2}(p_0 - p_{-1}) \sum_{\text{pairs}} \max(\mu_k, \mu_{\bar{k}})$$
[0.360]. If the sparser packet is favoured instead, the bound is [0.261]. The bound for saturating all I_- transitions with no overlap at all, $Q_0 + \frac{3}{2}(p_0 - p_{-1})$ [0.424], is unreachable.

4. **Inversion.** AFP gains are twice the saturation gains at the same spectral position, Eq. (13). Two-window AFP gives [0.298 at $\epsilon = 1$; 0.278 at $\epsilon = 0.8$], independent of the unknown RF selectivity.

5. **Per-spin selectivity sets most of the single-burn range** [0.232–0.272 over plausible to extreme selectivity]. Refilling strength and time-shaping add less [$\lesssim 0.009$]. Site resolution and the dipolar reservoir change single-burn optima by less than 10^{-3}.

6. **Steady state with continuous DNP.** DNP restores P while a weak burn converts it to tensor order. The optimal constant burn sits where only the I_- line exists (beyond the C–D edge). It gives [$Q = 0.202$–0.213, depending on the transport closure]. Burning the horn at the measured DNP rate lowers Q [0.175], because DNP is too slow (435 s) to replace P.

7. **Transport type is decisive.** If spectral transport stayed within each branch, the I_- saturation produced in the I_--only region would be carried into the overlap region while DNP resupplies P [steady Q up to 0.24–0.30]. With cross-branch transport of comparable strength, as the equal matrix elements of Sec. 7 suggest for uncorrelated pairs, stronger transport *lowers* Q [0.18 and 0.15 at 10× and 100×]. This mechanism is the largest open lever, and it is constrained by measurable refill behaviour.

8. **Timing.** For incoherent saturation the response to RF dose is concave: already-equalized transitions gain nothing from more power. Spreading a given dose evenly in time is optimal. Pulse trains, sweeps and optimized periodic schedules did not beat a steady burn [gain $\le$ 0.0016; long gaps lose up to 0.024]. A transient overshoot after RF switch-on [0.003–0.009] can be used once. *This conclusion does not extend to coherent high-power sequences*, which can invert populations.

12 What is known and what is not

Mechanism	Theory	d-butanol status	Measurement that would pin it
Pake spectrum, sites	established [3, 5]	fitted (0.8%)	–
Weak-drive RF rate	established [1]	strength fitted to one burn	B_1 calibration; power/duration series
Per-spin selectivity ($T_2(\Omega)$)	mechanisms known [6]	**unknown**	narrow-hole widths and initial rates at horn vs pedestal; mirror response
B_1 inhomogeneity	known	**unknown**	nutation; saturation-curve shape
AFP efficiency	established [22]	**unknown** for selective windows	selective sweeps with lineshape analysis
T_1: W_1, W_2	established [1]	**not separated**	relaxation of P and Q after manipulation
Same-branch diffusion (K, width)	established [17]	weakly identified	refill after narrow burns
Cross-branch ratio r_X	expected ~ 1 (equal matrix elements)	**never fitted**	refill of I_+ vs I_- after a burn in the I_--only region
DQ exchange	second order, weak	effective channel fitted strong	refill far from the burn
Nuclear dipolar reservoir	established [9, 10]	**unknown** C_d	hole asymmetry; off-resonance RF
DNP mechanism and selectivity	established [11, 21]	effective rate fitted	frequency-resolved build-up after burns
Microwave-modified transport	observed for protons [29]	**unknown**	RF burn/refill with microwaves on vs off

Open experimental puzzles.

1. **The persistent $\sim 20\%$ RF-on peak residual.** At about 0.1 W the peak falls quickly, then stalls near 20% under continued irradiation, while the pedestal burns to about zero; higher power burns the peak further. No fitted rate model reproduces this. Candidate explanations:
   - spatial B_1 inhomogeneity (a weakly driven fraction);
   - strong replenishment of the horn from dense same-branch partners;
   - hyperfine-shifted or otherwise detuned sub-ensembles;
   - rotating-frame or dipolar-reservoir saturation, which is not pure equalization;
   - RF heating.

   Plateau duration and power dependence would discriminate between them.
2. **Microwave on/off power ratio** (about 3×). This requires separating DNP pumping from microwave-modified transport.
3. **Deviations from equal spin temperature** between P and Q under DNP in ND_3 [31]. This tests whether quadrupolar order has its own temperature or its own contact to the electrons.

A Spin-1 operators

In the basis $(|+1\rangle, |0\rangle, |-1\rangle)$:

$$I_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad I_+ = \sqrt{2} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad I_x = \tfrac{1}{2}(I_+ + I_-), \quad 3I_z^2 - 2 = \mathrm{diag}(1, -2, 1).$$

The tensor polarization is $Q = \langle 3I_z^2 - 2 \rangle$ and the vector polarization is $P = \langle I_z \rangle$.

B Pake Jacobian in frequency coordinates

With $\omega = \Omega P_2(\cos\theta)$, we have $\cos^2\theta = (1 + 2\omega/\Omega)/3$ and $\sin^2\theta = \frac{2}{3}(1 - \omega/\Omega)$. The Jacobian is

$$J = |d\omega/d\theta| = \Omega\sqrt{2(1 - \omega/\Omega)(1 + 2\omega/\Omega)},$$

and the density is $\mu \propto \sin\theta/J \propto (1 + 2\omega/\Omega)^{-1/2}$, consistent with Eq. (6). Equation (18) then follows from the Laplace–Beltrami form, with no-flux boundaries at $\omega = -\Omega/2$ and $\omega = \Omega$.

C Rate-equation matrices

For one packet, write $u_\pm$ for the rate that raises m across the $\mathrm{I}_\pm$ transition and $d_\pm$ for the rate that lowers it:

$$\boldsymbol{R} = \begin{pmatrix} -d_+ & u_+ & 0 \\ d_+ & -u_+ - d_- & u_- \\ 0 & d_- & -u_- \end{pmatrix},$$

For RF, $u_\pm = d_\pm = g_\pm$; for DNP, $u \neq d$. The relaxation part to the lattice is

$$\boldsymbol{R}_{T_1} = \begin{pmatrix} -W_1 - W_2 & W_1 & W_2 \\ W_1 & -2W_1 & W_1 \\ W_2 & W_1 & -W_1 - W_2 \end{pmatrix}$$

(high-temperature form; detailed-balance factors $e^{\pm\beta_L\hbar\omega}$ multiply the upward rates).

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